ASSIGNMENT : 15

TITLE : Write a Java program to demonstrate exception handling using finally, throw, and throws.

Problem Statement : Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives : To implement finally, throw, and throws for effective exception management.

Theory : The finally block always executes regardless of whether an exception occurs, making it suitable for resource cleanup. The throw keyword explicitly generates an exception, while throws declares exceptions that a method may pass to its caller. These mechanisms improve structured error handling and application stability.

Exercise :

1. Create a Login program that throws an exception for invalid password and uses finally block.

java15a.java

java15a.java > Java > java15a > main(String[] args)

```java
import java.util.Scanner;

public class java15a {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        String correctPassword = "java123";

        try {
            System.out.print(s: "Enter Password: ");
            String password = sc.nextLine();
            sc.close();

            if (!password.equals(correctPassword)) {
                throw new Exception(message: "Invalid Password!");
            }

            System.out.println(x: "Login Successful.");

        } catch (Exception e) {
            System.out.println(e.getMessage());
        } finally {
            System.out.println(x: "Login process completed.");
        }
    }
}
```

OUTPUT :

- ```
  Enter Password: anu1234
  Invalid Password!
  Login process completed.
  ```

- ```
  Enter Password: java123
  Login Successful.
  Login process completed.
  ```

2. Create an **ATM PIN Verification** program that throws an exception for an invalid PIN entered by the user and uses a **finally** block to display a message indicating that the verification process has completed.

java15a.java | java15b.java

java15b.java > Java > java15b > main(String[] args)

```java
import java.util.Scanner;

public class java15b {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        int correctPin = 8222;

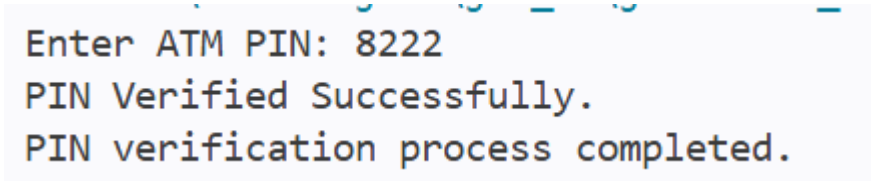
        try {
            System.out.print(s: "Enter ATM PIN: ");
            int pin = sc.nextInt();
            sc.close();

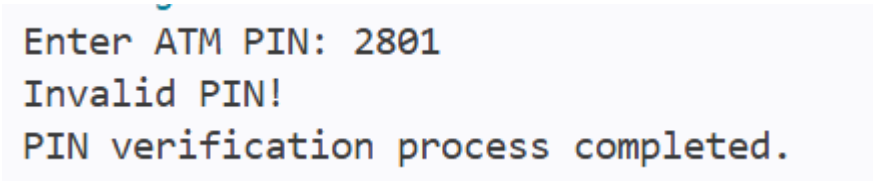
            if (pin != correctPin) {
                throw new Exception(message: "Invalid PIN!");
            }

            System.out.println(x: "PIN Verified Successfully.");

        } catch (Exception e) {
            System.out.println(e.getMessage());
        } finally {
            System.out.println(x: "PIN verification process completed.");
        }
    }
}
```

OUTPUT :

- 

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Outcome :

- Understood the use of **finally, throw,** and **throws** in Java exception handling.
- Implemented programs that generate, propagate, and handle exceptions effectively.
- Learned how these mechanisms improve structured error handling and resource management.

Conclusion :

This assignment demonstrated the use of **finally, throw,** and **throws** for effective exception handling in Java. These features help manage errors systematically, ensure proper resource cleanup, and improve the stability and maintainability of Java applications.